Protein-Glycan Interaction Resource at the National Center for Functional Glycomics (NCFG)

Project Number
5R24GM137763-04

Contact PI/Project Leader
CUMMINGS, RICHARD D

Awardee Organization
BETH ISRAEL DEACONESS MEDICAL CENTER

Description

Abstract Text

Abstract/Summary This R24 application for an NIGMS National and Regional Resource is to enable researchers in the under- served area of glycosciences and whose research needs glycoscience expertise, to have continued and improved access to state-of-the-art technologies to advance biomedical research and human health involving protein-glycan interactions and glycan recognition. Glycosylation is the most common and varied post- translational modification (PTM) in all living things, and each cell and organism generates unique PTMs and also glycolipids. Such resources proposed here are neither available to individual laboratories, nor are these specific technologies available commercially. With a base of ~7,000 users (83.6% new users) and hundreds of laboratories utilizing NCFG databases and resources, respectively, related to functional glycomics, our resource is obviously well used and represents a unique resource democratically accessible to all biomedical researchers. The increasing demand is evidenced by the nearly 10,000 individual, different glycan microarrays used at the NCFG resource center in just the last 4 years. Our emphasis on protein-glycan interactions is timely because research continues to uncover clues that these interactions are key to understanding the expression and functions of glycans in biological systems and their recognition by antibodies, glycan-binding proteins (GBPs) and lectins, in human and animal systems, and by microbial pathogens and gut microbiome. Our resource makes available an incredible variety of glycan microarray technologies, glycan libraries, and multiple databases not available elsewhere. The success of our resource has resulted in over a hundred publications by users in the past 5 years. Because of its success and rapidly increasing number of requests, and because we are the only resource of this kind available in the glycosciences, we propose three Specific Aims to continue making these invaluable resources accessible. Aim 1- We will continue to generate and improve defined glycan, glycopeptide, glycoprotein, and glycolipid microarrays, along with Luminex-based glycan-bead technologies, all unique to the resource. Aim 2- We will make available and further improve natural bifunctional fluorescent-tags (BFTs) originally pioneered by the NCFG, and will be utilized in glycan microarray and bead conjugations. Such novel BFTs, pioneered by the NCFG, give us advanced capabilities to label glycans in complex mixtures, purify them, and then quantify and quantitatively print microarrays at varying densities. Aim 3- We will continue to use BFT technology to produce and improve natural glycan microarrays from many sources of cells, tissues, and glycoconjugates, comprising a unique and valuable tool for the resource center. Thus, our Protein-Glycan Interaction Resource provides unique capabilities in glycan- binding expertise and technologies, teaching opportunities, and a wide range of services and outreach to an emerging field rapidly becoming recognized as a key factor in health and disease.

Public Health Relevance Statement

Narrative The proposed request for resource support is to allow continued access for thousands of users to the state-of- the-art technologies important in defining protein-glycan interactions, which are necessary for understanding the expression and functions of glycans in biological systems in health and disease, and glycan recognition by antibodies, glycan-binding proteins (GBPs), and lectins. To date, over 7,000 users and hundreds of laboratories have utilized the NCFG databases and resources, seeking information and assistance related to Functional Glycomics and our Resource research data has led them to explore glycosciences. This Resource provides unique and necessary capabilities in technology, teaching, and service related to understanding protein-glycan interactions to a broad swath of researchers that are experts and non-experts in the field of glycosciences.

Project Terms

Animals | Antibodies | Area | Autoimmunity | Bacteria | Binding | Binding Proteins | Biomedical Research | Cells | Collaborations | Communicable Diseases | Communities | Complex Mixtures | DNA Library | Data | Databases | Derivation procedure | Development | Disease

Read Less

Educational process of instructing | Ensure | Fluorescence | Funding | Generations | Glycobiology | Glycoconjugates | Glycolipids | Glycopeptides | Glycoproteins | Glycosaminoglycans | Health | High Pressure Liquid Chromatography | Human | Human Development | Immunology | Improve Access | Individual | Journals | Label | Laboratories | Lead | Lectin | Libraries | Malignant Neoplasms | Manuscripts | Microarray Analysis | Microbe | Modification | Molecular Conformation | Monitor | National Institute of General Medical Sciences | Natural Resources | Natural Source | Oligosaccharides | Organism | Outcome | Peer Review | Peptide Library | Polysaccharides | Post-Translational Protein Processing | Printing | Production | Proteins | Publications | Publishing | R24 | RNA library | Recombinants | Research | Research Personnel | Resources | Services | Shotguns | Source | Sulfate | System | Technology | Tissues | Virus | animal data | base | biological systems | cell type | density | glycosylation | gut microbiome | improved | innovation | inorganic phosphate | microbial | new technology | novel | outreach | pathogenic microbe | preservation | programs | success | tool | underserved area | web site

Details

Contact PI/ Project Leader

Name
CUMMINGS, RICHARD D

Title
PROFESSOR OF SURGERY

Contact
View Email

Other PIs

Not Applicable

Program Official

Name
BOND, MICHELLE RUEFFER

Contact
View Email

Organization

Name
BETH ISRAEL DEACONESS MEDICAL CENTER

City
BOSTON

Country
UNITED STATES (US)

Department Type
Unavailable

Organization Type
Independent Hospitals

State Code
MA

Congressional District
07

Other Information

Opportunity Number
PAR-19-301

Study Section
ZGM1-BBCB-4(NR)

Fiscal Year
2023

Award Notice Date
26-June-2023

Administering Institutes or Centers
National Institute of General Medical Sciences

CFDA Code
859

DUNS Number
071723621

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Project Start Date **01-July-2020**

Project End Date **30-June-2025**

Budget Start Date **01-July-2023**

Budget End Date **30-June-2024**

Project Funding Information for 2023

Total Funding
$962,500

Direct Costs
$550,000

Indirect Costs
$412,500

Year	Funding IC	FY Total Cost by IC

Sub Projects

No Sub Projects information available for 5R24GM137763-04

Publications

Export Disclaimer

Journal (Link to PubMed abstract)	Authors	Publication Year	Similar Publications	CitedBy
Structure-function correlates of fibrinogen binding by Acinetobacter adhesins critical in catheter-associated urinary tract infections.				
Proceedings of the National Academy of Sciences of the United States of America 2023 Jan 24; 120 (4) e2212694120	Tamadonfar, Kevin O; Di Venanzio, Gisela; Pinkner, Jerome S; Dodson, Karen W; Kalas, Vasilios; Zimmerman, Maxwell I; Bazan Villicana, Jesus; Bowman, Gregory R; Feldman, Mario F; Hultgren, Scott J View Less	2023		
ABO blood group antigens and differential glycan expression: Perspective on the evolution of common human enzyme deficiencies.				
iScience 2023 Jan 20; 26 (1) 105798	Jajosky, Ryan Philip; Wu, Shang-Chuen; Zheng, Leon; Jajosky, Audrey N; Jajosky, Philip G; Josephson, Cassandra D; Hollenhorst, Marie A; Sackstein, Robert; Cummings, Richard D; Arthur, Connie M; Stowell, Sean R View Less	2023		
Heparan sulfate glycomimetics via iterative assembly of "clickable" disaccharides.				
Chemical science 2023 Mar 29; 14 (13) 3514-3522	Yang, Cangjie; Deng, Yu; Wang, Yang; Xia, Chaoshuang; Mehta, Akul Y; Baker, Kelly J; Samal, Anuj; Booneimsri, Putthipong; Lertmaneedang, Chanthakarn; Hwang, Seung; Flynn, James P; Cao, Muqing; Liu, Chao; Zhu, Alec C; Cummings, Richard D; Lin, Cheng; Mohanty, Udayan; Niu, Jia View Less	2023		
Longitudinal gut microbiome analyses and blooms of pathogenic strains during lupus disease flares.				
Annals of the rheumatic diseases 2023 Oct; 82 (10) 1315-1327	Azzouz, Doua F; Chen, Ze; Izmirly, Peter M; Chen, Lea Ann; Li, Zhi; Zhang, Chongda; Mieles, David; Trujillo, Kate; Heguy, Adriana; Pironti, Alejandro; Putzel, Greg G; Schwudke, Dominik; Fenyo, David; Buyon, Jill P; Alekseyenko, Alexander V; Gisch, Nicolas; Silverman, Gregg J View Less	2023		
Sialylation regulates neutrophil transepithelial migration, CD11b/CD18 activation, and intestinal mucosal inflammatory function.				
JCI insight 2023 Mar 08; 8 (5)	Azcutia, Veronica; Kelm, Matthias; Fink, Dylan; Cummings, Richard D; Nusrat, Asma; Parkos, Charles A; Brazil, Jennifer C	2023		
Novel banana lectin CAR-T cells to target pancreatic tumors and tumor-associated stroma.				

Journal (Link to PubMed abstract)	Authors	Publication Year	Similar Publications	CitedBy	iCite RCR
Journal for immunotherapy of cancer 2023 Jan; 11 (1)	McKenna, Mary K; Ozcan, Ada; Brenner, Daniel; Watanabe, Norihiro; Legendre, Maureen; Thomas, Dafydd G; Ashwood, Christopher; Cummings, Richard D; Bonifant, Challice; Markovitz, David M; Brenner, Malcolm K View Less	2023			

Targeting altered glycosylation in secreted tumor glycoproteins for broad cancer detection.

Glycobiology 2023 Aug 14; 33 (7) 567-578	Kudelka, Matthew R; Gu, Wei; Matsumoto, Yasuyuki; Ju, Tongzhong; Barnes Ii, Richard H; Kardish, Robert J; Heimburg-Molinaro, Jamie; Lehoux, Sylvain; Zeng, Junwei; Cohen, Cynthia; Robinson, Brian S; Shah, Kinjal S; Chaikof, Elliot L; Stowell, Sean R; Cummings, Richard D View Less	2023			

2019-20 H1N1 clade A5a.1 viruses have better in vitro replication compared with the co-circulating A5a.2 clade.

bioRxiv : the preprint server for biology 2023 Feb 26;	Swanson, Nicholas J; Marinho, Paula; Dziedzic, Amanda; Jedlicka, Anne; Liu, Hsuan; Fenstermacher, Katherine; Rothman, Richard; Pekosz, Andrew	2023			

Glycosphingolipids in human parasites.

FEBS open bio 2023 Sep; 13 (9) 1625-1635	Cummings, Richard D	2023			

Functional HIV-1/HCV cross-reactive antibodies isolated from a chronically co-infected donor.

Cell reports 2023 Feb 28; 42 (2) 112044	Pilewski, Kelsey A; Wall, Steven; Richardson, Simone I; Manamela, Nelia P; Clark, Kaitlyn; Hermanus, Tandile; Binshtein, Elad; Venkat, Rohit; Sautto, Giuseppe A; Kramer, Kevin J; Shiakolas, Andrea R; Setliff, Ian; Salas, Jordan; Mapengo, Rutendo E; Suryadevara, Naveen; Brannon, John R; Beebout, Connor J; Parks, Rob; Raju, Nagarajan; Frumento, Nicole; Walker, Lauren M; Fechter, Emilee Friedman; Qin, Juliana S; Murji, Amyn A; Janowska, Katarzyna; Thakur, Bhishem; Lindenberger, Jared; May, Aaron J; Huang, Xiao; Sammour, Salam; Acharya, Priyamvada; Carnahan, Robert H; Ross, Ted M; Haynes, Barton F; Hadjifrangiskou, Maria; Crowe Jr, James E; Bailey, Justin R; Kalams, Spyros; Morris, Lynn; Georgiev, Ivelin S View Less	2023			

Whole microbe arrays accurately predict interactions and overall antimicrobial activity of galectin-8 toward distinct strains of Streptococcus pneumoniae.

Scientific reports 2023 Apr 01; 13 (1) 5324	Wu, Shang-Chuen; Jan, Hau-Ming; Vallecillo-Zúniga, Mary L; Rathgeber, Matthew F; Stowell, Caleb S; Murdock, Kaleb L; Patel, Kashyap R; Nakahara, Hirotomo; Stowell, Carter J; Nahm, Moon H; Arthur, Connie M; Cummings, Richard D; Stowell, Sean R View Less	2023			

Journal (Link to PubMed abstract)	Authors	Publication Year	Similar Publications	CitedBy	iCite RCR
N-glycans show distinct spatial distribution in mouse brain.					
bioRxiv : the preprint server for biology 2023 May 30;	Noel, Maxence; Cummings, Richard D; Mealer, Robert G	2023			
Large-Scale and Site-Specific Mapping of the Murine Brain O-Glycoproteome with IMPa.					
Analytical chemistry 2023 Sep 12; 95 (36) 13423-13430	Suttapitugsakul, Suttipong; Matsumoto, Yasuyuki; Aryal, Rajindra P; Cummings, Richard D	2023			
The Inflammation Biomarker GlycA Reflects Plasma N-Glycan Branching.					
Clinical chemistry 2023 Jan 04; 69 (1) 80-87	Noel, Maxence; Chasman, Daniel I; Mora, Samia; Otvos, James D; Palmer, Christopher D; Parsons, Patrick J; Smoller, Jordan W; Cummings, Richard D; Mealer, Robert G View Less	2023			
Increased Galectin-9 Levels Correlate with Disease Activity in Patients with DMARD-Naïve Rheumatoid Arthritis and Modulate the Secretion of MCP-1 and IL-6 from Synovial Fibroblasts.					
Cells 2023 Jan 15; 12 (2)	Nielsen, Morten A; Køster, Ditte; Mehta, Akul Y; Stengaard-Pedersen, Kristian; Busson, Pierre; Junker, Peter; Hørslev-Petersen, Kim; Hetland, Merete Lund; Østergaard, Mikkel; Hvid, Malene; Leffler, Hakon; Kragstrup, Tue W; Cummings, Richard D; Deleuran, Bent View Less	2023			
Anti-schistosomal immunity to core xylose/fucose in N-glycans.					
Frontiers in molecular biosciences 2023; 10 1142620	Prasanphanich, Nina Salinger; Leon, Kristoffer; Secor, W Evan; Shoemaker, Charles B; Heimburg-Molinaro, Jamie; Cummings, Richard D	2023			
Chemokine binding to PSGL-1 is controlled by O-glycosylation and tyrosine sulfation.					
Cell chemical biology 2023 Aug 17; 30 (8) 893-905.e7	Goth, Christoffer K; Mehta, Akul Y; McQuillan, Alyssa M; Baker, Kelly J; Hanes, Melinda S; Park, Simon S; Stavenhagen, Kathrin; Hjortø, Gertrud M; Heimburg-Molinaro, Jamie; Chaikof, Elliot L; Rosenkilde, Mette M; Cummings, Richard D View Less	2023			
Mucin O-glycans are natural inhibitors of Candida albicans pathogenicity.					
Nature chemical biology 2022 Jul; 18 (7) 762-773	Takagi, Julie; Aoki, Kazuhiro; Turner, Bradley S; Lamont, Sabrina; Lehoux, Sylvain; Kavanaugh, Nicole; Gulati, Megha; Valle Arevalo, Ashley; Lawrence, Travis J; Kim, Colin Y; Bakshi, Bhavya; Ishihara, Mayumi; Nobile, Clarissa J; Cummings, Richard D; Wozniak, Daniel J; Tiemeyer, Michael; Hevey, Rachel; Ribbeck, Katharina View Less	2022			iCite 3.54
Computationally guided conversion of the specificity of E-selectin to mimic that of Siglec-8.					
Proceedings of the National Academy of Sciences of the United States of America 2022 Oct 11; 119 (41) e2117743119	Wang, Xiaocong; Hanes, Melinda S; Cummings, Richard D; Woods, Robert J	2022			

Journal (Link to PubMed abstract)	Authors	Publication Year	Similar Publications	CitedBy	iCite RCR
N-glycosylation of mannose receptor (CD206) regulates glycan binding by C-type lectin domains.					
The Journal of biological chemistry 2022 Dec; 298 (12) 102591	Stavenhagen, Kathrin; Mehta, Akul Y; Laan, Lisa; Gao, Chao; Heimburg-Molinaro, Jamie; van Die, Irma; Cummings, Richard D	2022			
Sialylated and sulfated N-Glycans in MDCK and engineered MDCK cells for influenza virus studies.					
Scientific reports 2022 Jul 26; 12 (1) 12757	Byrd-Leotis, Lauren; Jia, Nan; Matsumoto, Yasuyuki; Lu, Dongli; Kawaoka, Yoshihiro; Steinhauer, David A; Cummings, Richard D	2022			
Structural analysis of N-glycans in chicken trachea and lung reveals potential receptors of chicken influenza viruses.					
Scientific reports 2022 Feb 08; 12 (1) 2081	Suzuki, Noriko; Abe, Tatsuya; Natsuka, Shunji	2022			iCite 1.77
The mannose receptor ligands and the macrophage glycome.					
Current opinion in structural biology 2022 Aug; 75 102394	Cummings, Richard D	2022			iCite 1.66
A Useful Guide to Lectin Binding: Machine-Learning Directed Annotation of 57 Unique Lectin Specificities.					
ACS chemical biology 2022 Nov 18; 17 (11) 2993-3012	Bojar, Daniel; Meche, Lawrence; Meng, Guanmin; Eng, William; Smith, David F; Cummings, Richard D; Mahal, Lara K	2022			iCite 15.21
The schizophrenia-associated variant in SLC39A8 alters protein glycosylation in the mouse brain.					
Molecular psychiatry 2022 Mar; 27 (3) 1405-1415	Mealer, Robert G; Williams, Sarah E; Noel, Maxence; Yang, Bo; D'Souza, Alexandria K; Nakata, Toru; Graham, Daniel B; Creasey, Elizabeth A; Cetinbas, Murat; Sadreyev, Ruslan I; Scolnick, Edward M; Woo, Christina M; Smoller, Jordan W; Xavier, Ramnik J; Cummings, Richard D View Less	2022			iCite 2.57
Use of a glycomics array to establish the anti-carbohydrate antibody repertoire in type 1 diabetes.					
Nature communications 2022 Nov 01; 13 (1) 6527	Tran, Paul M H; Dong, Fran; Kim, Eileen; Richardson, Katherine P; Tran, Lynn K H; Waugh, Kathleen; Hopkins, Diane; Cummings, Richard D; Wang, Peng George; Rewers, Marian J; She, Jin-Xiong; Purohit, Sharad View Less	2022			
Differential recognition of oligomannose isomers by glycan-binding proteins involved in innate and adaptive immunity.					
Science advances 2021 06; 7 (24)	Gao, Chao; Stavenhagen, Kathrin; Eckmair, Barbara; McKitrick, Tanya R; Mehta, Akul Y; Matsumoto, Yasuyuki; McQuillan, Alyssa M; Hanes, Melinda S; Eris, Deniz; Baker, Kelly J; Jia, Nan; Wei, Mohui; Heimburg-Molinaro, Jamie; Ernst, Beat; Cummings, Richard D View Less	2021			iCite 1.84
mAb Das-1 recognizes 3'-Sulfated Lewis A/C, which is aberrantly expressed during metaplastic and oncogenic transformation of several gastrointestinal Epithelia.					
PloS one 2021; 16 (12) e0261082	Brown, Jeffrey W; Das, Koushik K; Kalas, Vasilios; Das, Kiron M; Mills, Jason C	2021			iCite 0.61
Tumor cells express pauci- and oligomannosidic N-glycans in glycoproteins recognized by the mannose receptor (CD206).					

Journal (Link to PubMed abstract)	Authors	Publication Year	Similar Publications	CitedBy	iCite RCR
Cellular and molecular life sciences : CMLS 2021 Jul; 78 (14) 5569-5585	Stavenhagen, Kathrin; Laan, Lisa C; Gao, Chao; Mehta, Akul Y; Heimburg-Molinaro, Jamie; Glickman, Jonathan N; van Die, Irma; Cummings, Richard D	2021			1.27

An Egg-Derived Sulfated N-Acetyllactosamine Glycan Is an Antigenic Decoy of Influenza Virus Vaccines.

mBio 2021 06 29; 12 (3) e0083821	Guthmiller, Jenna J; Utset, Henry A; Henry, Carole; Li, Lei; Zheng, Nai-Ying; Sun, Weina; Costa Vieira, Marcos; Zost, Seth; Huang, Min; Hensley, Scott E; Cobey, Sarah; Palese, Peter; Wilson, Patrick C View Less	2021			0.63

Influenza vaccination in the elderly boosts antibodies against conserved viral proteins and egg-produced glycans.

The Journal of clinical investigation 2021 Jul 01; 131 (13)	Jung, Jiwon; Mundle, Sophia T; Ustyugova, Irina V; Horton, Andrew P; Boutz, Daniel R; Pougatcheva, Svetlana; Prabakaran, Ponraj; McDaniel, Jonathan R; King, Gregory R; Park, Daechan; Person, Maria D; Ye, Congxi; Tan, Bing; Tanno, Yuri; Kim, Jin Eyun; Curtis, Nicholas C; DiNapoli, Joshua; Delagrave, Simon; Ross, Ted M; Ippolito, Gregory C; Kleanthous, Harry; Lee, Jiwon; Georgiou, George View Less	2021			0.89

SARS-CoV-2 and other coronaviruses bind to phosphorylated glycans from the human lung.

Virology 2021 10; 562 142-148	Byrd-Leotis, Lauren; Lasanajak, Yi; Bowen, Thomas; Baker, Kelly; Song, Xuezheng; Suthar, Mehul S; Cummings, Richard D; Steinhauer, David A	2021			1.40

Major differences in glycosylation and fucosyltransferase expression in low-grade versus high-grade bladder cancer cell lines.

Glycobiology 2021 12 18; 31 (11) 1444-1463	Ezeabikwa, Bernadette; Mondal, Nandini; Antonopoulos, Aristotelis; Haslam, Stuart M; Matsumoto, Yasuyuki; Martin-Caraballo, Miguel; Lehoux, Sylvain; Mandalasi, Msano; Ishaque, Ali; Heimburg-Molinaro, Jamie; Cummings, Richard D; Nyame, Anthony K View Less	2021			0.91

CRISPR-based peptide library display and programmable microarray self-assembly for rapid quantitative protein binding assays.

Molecular cell 2021 09 02; 81 (17) 3650-3658.e5	Barber, Karl W; Shrock, Ellen; Elledge, Stephen J	2021			0.90

Novel lamprey antibody recognizes terminal sulfated galactose epitopes on mammalian glycoproteins.

Communications biology 2021 06 03; 4 (1) 674	McKitrick, Tanya R; Bernard, Steffen M; Noll, Alexander J; Collins, Bernard C; Goth, Christoffer K; McQuillan, Alyssa M; Heimburg-Molinaro, Jamie; Herrin, Brantley R; Wilson, Ian A; Cooper, Max D; Cummings, Richard D View Less	2021			1.74

Proteomic analysis of the host-pathogen interface in experimental cholera.

Journal (Link to PubMed abstract)	Authors	Publication Year	Similar Publications	CitedBy	iCite RCR
Nature chemical biology 2021 Nov; 17 (11) 1199-1208	Zoued, Abdelrahim; Zhang, Hailong; Zhang, Ting; Giorgio, Rachel T; Kuehl, Carole J; Fakoya, Bolutife; Sit, Brandon; Waldor, Matthew K	2021			iCite 0.90
PDX-derived organoids model in vivo drug response and secrete biomarkers.					
JCI insight 2020 11 05; 5 (21)	Huang, Ling; Bockorny, Bruno; Paul, Indranil; Akshinthala, Dipikaa; Frappart, Pierre-Oliver; Gandarilla, Omar; Bose, Arindam; Sanchez-Gonzalez, Veronica; Rouse, Emily E; Lehoux, Sylvain D; Pandell, Nicole; Lim, Christine M; Clohessy, John G; Grossman, Joseph; Gonzalez, Raul; Del Pino, Sofia Perea; Daaboul, George; Sawhney, Mandeep S; Freedman, Steven D; Kleger, Alexander; Cummings, Richard D; Emili, Andrew; Muthuswamy, Lakshmi B; Hidalgo, Manuel; Muthuswamy, Senthil K View Less	2020			iCite 3.89
Tools for generating and analyzing glycan microarray data.					
Beilstein journal of organic chemistry 2020; 16 2260-2271	Mehta, Akul Y; Heimburg-Molinaro, Jamie; Cummings, Richard D	2020			iCite 0.90
Intestinal epithelial glycosylation in homeostasis and gut microbiota interactions in IBD.					
Nature reviews. Gastroenterology & hepatology 2020 Oct; 17 (10) 597-617	Kudelka, Matthew R; Stowell, Sean R; Cummings, Richard D; Neish, Andrew S	2020			iCite 7.00
GlycoGlyph: a glycan visualizing, drawing and naming application.					
Bioinformatics (Oxford, England) 2020 06 01; 36 (11) 3613-3614	Mehta, Akul Y; Cummings, Richard D	2020			iCite 2.41
Enhanced Pro-apoptotic Effects of Fe(II)-Modified IVIG on Human Neutrophils.					
Frontiers in immunology 2020; 11 973	Graeter, Stefanie; Schneider, Christoph; Verschoor, Daniëlle; von Däniken, Sandro; Seibold, Frank; Yawalkar, Nikhil; Villiger, Peter; Dimitrov, Jordan D; Smith, David F; Cummings, Richard D; Simon, Hans-Uwe; Vassilev, Tchavdar; von Gunten, Stephan View Less	2020			iCite 0.32
Antibodies from Lampreys as Smart Anti-Glycan Reagents (SAGRs): Perspectives on Their Specificity, Structure, and Glyco-genomics.					
Biochemistry 2020 09 01; 59 (34) 3111-3122	McKitrick, Tanya R; Eris, Deniz; Mondal, Nandini; Aryal, Rajindra P; McCurley, Nathanael; Heimburg-Molinaro, Jamie; Cummings, Richard D	2020			iCite 1.31
Parallel Glyco-SPOT Synthesis of Glycopeptide Libraries.					
Cell chemical biology 2020 09 17; 27 (9) 1207-1219.e9	Mehta, Akul Y; Veeraiah, Ravi Kumar H; Dutta, Sucharita; Goth, Christoffer K; Hanes, Melinda S; Gao, Chao; Stavenhagen, Kathrin; Kardish, Robert; Matsumoto, Yasuyuki; Heimburg-Molinaro, Jamie; Boyce, Michael; Pohl, Nicola L B; Cummings, Richard D View Less	2020			iCite 0.81

Journal (Link to PubMed abstract)	Authors	Publication Year	Similar Publications	CitedBy	iCite RCR
Unique repertoire of anti-carbohydrate antibodies in individual human serum.					
Scientific reports 2020 09 22; 10 (1) 15436	Luetscher, Ralph N D; McKitrick, Tanya R; Gao, Chao; Mehta, Akul Y; McQuillan, Alyssa M; Kardish, Robert; Boligan, Kayluz Frias; Song, Xuezheng; Lu, Lenette; Heimburg-Molinaro, Jamie; von Gunten, Stephan; Alter, Galit; Cummings, Richard D View Less	2020			iCite 1.17

Patents

Export

Patent Number	Patent Title	Patent Owner	Primary Agency
11766481	GLYCAN MODIFIED SHORT INTERFERING RNA	BOSTON CHILDREN'S HOSPITAL	NIH

Outcomes

The Project Outcomes shown here are displayed verbatim as submitted by the Principal Investigator (PI) for this award. Any opinions, findings, and conclusions or recommendations expressed are those of the PI and do not necessarily reflect the views of the National Institutes of Health. NIH has not endorsed the content below.

No Outcomes available for 5R24GM137763-04

Clinical Studies

No Clinical Studies information available for 5R24GM137763-04

News and More

Related News Releases

News	Journal Article	PubMed Abstract	Relea
Vaccines grown in eggs induce antibody response against an egg-associated glycan		July 2021	

History

No Historical information available for 5R24GM137763-04

Similar Projects

No Similar Projects information available for 5R24GM137763-04